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An Agentic Artificial Intelligence System for Crop Advising via Multimodal Data Integration and Language Model Analysis

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Abstract

Choosing which crop to cultivate is one of the most consequential decisions in agricultural planning. The choice depends on the physical properties of the land, on local climatic conditions and on how different crops have performed in the same area in previous seasons. Climate variability makes this decision increasingly difficult, since conditions that were stable for decades may change within a few years, affecting both the suitability of traditional crops and the opportunities offered by new ones.

Earth observation provides a large amount of information to support this decision. Satellite missions acquire frequent images of the entire planet, land-cover products describe how the land is used, climate reanalyses provide consistent records of temperature, precipitation and soil moisture, and national agencies publish detailed agricultural statistics. Most of these sources are openly accessible. However, they differ in modality (images, rasters, time series, tables), spatial and temporal resolution, and access mechanism. Turning them into actionable advice for a specific location requires retrieving the right data, integrating

them, and reasoning over the result.

The objective of this thesis is to design and evaluate an agentic system that, given a location in the contiguous United States, collects multimodal Earth observation and agricultural data and uses a Vision-Language Model to recommend the most suitable crop and a rotation crop, explaining the reasoning behind the recommendation. The remainder of this thesis is organised as follows. Chapter 2 reviews the state of the art in remote sensing image analysis, discusses the limitations of current approaches and presents the research question. Chapter 3 describes the architecture of the proposed system and its components. Chapter 4 presents the experimental environment, the datasets, the models and the evaluation strategy. Finally, Chapter 5 reports and discusses the results.

Chapter 1

Introduction

1.1 Remote sensing for crop advisory

Remote sensing is the acquisition of information about an object or an area without physical contact with it (Mårtensson, 2011). Although the term is commonly associated with satellites, the definition also covers aerial and ground-based sensors. The availability of satellite and airborne sensors has made remote sensing data abundant and diverse, and their analysis has become an active research field with applications ranging from environmental monitoring to precision agriculture.

Crop advisory can be regarded as a high-level task that relies on several lower-level ones. To recommend a crop, a system must first understand how the land is currently used, and then assess whether the climatic conditions of the area are favourable to each candidate crop. Two tasks are therefore particularly relevant.

1.1.1 Scene classification

Scene classification consists of assigning a single semantic label to an entire image. Unlike object detection or pixel-wise segmentation, it requires the model to capture the global context of the scene, and it is made difficult by the high variability within each class and the similarity between different classes (Huang and Hu, 2026). Recognising the type of scene, for instance cropland as opposed to urban areas or water, is a first step towards understanding a location, although crop advisory requires additional information that a single label cannot convey. The evolution of methods for this task is reviewed in Chapter 2.

1.1.2 Climate monitoring

Climate monitoring tracks variables such as temperature, precipitation, solar radiation and soil moisture over long periods, in order to characterise the climate of an area and detect its changes. This information is essential for agriculture: it allows farmers to maintain productive practices under changing conditions and, where possible, to benefit from them, for example by adopting crops that become suitable thanks to longer growing seasons (CTCN). For crop advisory, climate variables complement the information extracted from images by describing the conditions under which a crop will grow.

1.2 Remote sensing data

Remote sensing systems are usually divided into two families. *Passive* sensors measure the solar radiation reflected by the Earth's surface (or the energy it emits), and therefore depend on illumination and atmospheric conditions. *Active* sensors emit their own signal towards the target and measure the portion that is reflected back. The two families produce data with different characteristics and are suited to different applications.

1.2.1 Optical and multispectral imagery

Optical imagery is the oldest form of remote sensing and is acquired by passive sensors. Modern optical instruments split the reflected radiation into several spectral bands, each measured independently. The result is a multispectral image, which can be represented as a three-dimensional array whose dimensions correspond to height, width and number of bands. Each pixel stores a value proportional to the reflectance of the surface in a given band; images with hundreds of narrow, contiguous bands are referred to as hyperspectral.

Multispectral data are particularly valuable in agriculture because vegetation has a characteristic spectral signature. Bands in the red-edge and near-infrared regions are sensitive to chlorophyll content and canopy structure, and are commonly combined into vegetation indices that reflect crop health. The Sentinel-2 mission, used in this work,

provides multispectral images with spatial resolutions between 10 and 60 m and a revisit time of a few days (Drusch et al., 2012).

1.2.2 SAR and LiDAR

Synthetic Aperture Radar (SAR) and Light Detection and Ranging (LiDAR) are the most common active sensing techniques (Agrawal and Khairnar, 2019). SAR operates in the microwave range, at longer wavelengths than optical sensors; it can therefore acquire data through clouds and regardless of daylight, and its signal is sensitive to surface roughness and moisture. Each pixel of a SAR image is a complex number encoding the amplitude and phase of the backscattered signal. LiDAR emits laser pulses, typically from aircraft or drones, and measures their return time to build high-resolution three-dimensional models of the terrain and vegetation, used for instance for digital elevation models and forest mapping.

SAR and LiDAR data are not used in this thesis, which relies on optical imagery. SAR, in particular, is a natural extension for regions with persistent cloud cover.

1.2.3 Land-cover, climate and agricultural statistics

Besides raw imagery, crop advisory relies on derived and auxiliary data. Land-cover products assign a class to each pixel and describe how the

land is used, providing direct evidence of which crops are grown in an area. Climate datasets, as discussed above, describe the conditions under which crops grow. Finally, agricultural statistics report the yields actually obtained for each crop, linking environmental conditions to production outcomes. Combining these heterogeneous sources is one of the central challenges addressed in this work.

1.3 Artificial intelligence for remote sensing

Artificial intelligence is now widely used to analyse remote sensing data, owing to its ability to capture complex patterns in large datasets. Typical applications include land-cover change detection, urban expansion monitoring, crop health assessment and damage evaluation after natural disasters (Gu and Zeng, 2024). Three aspects of modern AI are particularly relevant to this thesis.

1.3.1 Representation learning

The performance of machine learning models depends heavily on how the input data are represented (Bengio et al., 2013). Traditional approaches relied on hand-crafted features designed by domain experts, a process that does not scale with the volume and variety of modern data. Representation learning automates this process by learning

features directly from the data. This shift has produced substantial improvements across many domains; in speech recognition, for example, learned representations reduced word error rates by about 30% relative to previous systems (Bengio et al., 2013). In remote sensing, learned representations underpin both deep classification networks and the large pretrained models discussed in Chapter 2.

1.3.2 Multimodal data integration

Real-world problems rarely involve a single type of data. Crop advisory, for instance, requires combining images, categorical land-cover information, climate time series and tabular statistics. Multimodal data integration aims to represent such heterogeneous data so that they can be processed jointly. Vision-Language Models (VLMs) offer a flexible solution to this problem: by combining a visual encoder with a large language model, they can process an image together with textual and structured information in a single input (Liu et al., 2023), and generate an answer in natural language.

1.3.3 Agentic AI

An AI agent is a system that perceives its environment, reasons about a goal, plans a sequence of actions and executes them, observing the results to decide the next step (Stryker; Wang et al., 2024a). In agents based on language models, the model acts as the reasoning component

and interacts with the environment by invoking external tools, such as search engines, databases or APIs, so that each observation informs the following decision (Yao et al., 2023). The environment plays a twofold role: it is the source from which the agent gathers information, and the space in which its actions take effect. This paradigm is well suited to problems such as crop advisory, where the information needed to answer a question is not available in advance but must be collected according to the goal.

1.4 The Information Retrieval perspective

Information Retrieval (IR) studies how to find, among large collections, the information that satisfies a user's need (Manning et al., 2008). Although it originated with statistical models of text, IR has progressively incorporated semantic and learned representations, and today provides general methods for accessing heterogeneous information.

Two IR techniques are particularly relevant to this thesis. *Crawling* systematically explores a collection of resources by following the links between them, deciding which resource to visit next and respecting the usage constraints of each host; focused crawlers restrict the exploration to resources relevant to a given goal (Chakrabarti et al., 1999). These principles apply to any collection that can be modelled as a graph, including a set of geographic locations. *Retrieval-augmented generation* (RAG) grounds the output of a language model in external information,

which is retrieved for each query and added to the model input (Lewis et al., 2020). Both techniques address the same need that arises in crop advisory: collecting the information relevant to a specific request and making it available to the component that produces the answer.

Chapter 2

Related Work

This chapter reviews the research areas on which this thesis builds. Section 2.1 summarises the evolution of methods for remote sensing image analysis, from hand-crafted features to foundation models. Sections 2.2 and 2.3 discuss vision-language models and agentic systems for Earth observation, which are the works closest to the proposed system. Section 2.4 reviews crop recommendation systems, Section 2.5 the Information Retrieval techniques underlying data collection, and Section 2.6 the evaluation of natural-language outputs. Section 2.7 positions this work with respect to the literature, Section 2.8 summarises the open problems, and Section 2.9 formulates the research questions.

2.1 Remote sensing image analysis

2.1.1 Hand-crafted features and classical machine learning

Early approaches to remote sensing scene classification relied on features designed by experts, such as texture descriptors computed with grey-level co-occurrence matrices and Gabor filters (Mirzapour and Ghassemian, 2015). These features were then fed to classical classifiers, including Support Vector Machines (Mountrakis et al., 2011) and ensemble methods such as Random Forests and boosting (Dou et al., 2024). While effective on small datasets, these pipelines generalise poorly across sensors and geographic regions, and manual feature design does not scale with the volume and resolution of modern imagery (Huang and Hu, 2026).

2.1.2 Deep learning

Convolutional Neural Networks (CNNs) replaced hand-crafted features with representations learned end-to-end. AlexNet was among the first deep architectures applied to high-resolution scene classification; Han et al. (Han et al., 2017) extended a pretrained AlexNet with pyramid pooling to capture features at multiple scales. CNNs, however, assume data arranged on a regular grid. Graph Neural Networks (GNNs) have been adopted for irregular structures, such as LiDAR point clouds or

networks of ground stations, where they model spatial relationships explicitly (Zhao et al., 2025). More recently, Vision Transformers (ViTs) (Dosovitskiy et al., 2021), which split an image into patches and process them with self-attention (Vaswani et al., 2017), have become the dominant backbone for visual recognition. ViTs are also the visual encoders of the vision-language models used in this thesis.

2.1.3 Foundation models and generative models

Foundation models are pretrained on large, often unlabelled, collections of multi-sensor data through self-supervised objectives, and are then adapted to downstream tasks with limited supervision, enabling few-shot and zero-shot transfer (Xiao et al., 2025). Generative models, in particular diffusion models, are used for image restoration, cloud removal, super-resolution and translation between sensors (Adamiak, 2023). Both families have improved performance on individual perception tasks. However, they produce dense representations or images rather than answers to user questions, and they are not designed to decide which data should be collected to solve a task.

2.2 Vision-language models for remote sensing

Vision-Language Models (VLMs) couple a visual encoder with a large language model through a connector that projects visual features into the embedding space of the language model (Liu et al., 2023). Instruction tuning on image–text pairs enables general-purpose VLMs, such as LLaVA (Liu et al., 2023) and Qwen2-VL (Wang et al., 2024b), to answer free-form questions about an image.

Several works adapt this paradigm to remote sensing. RSGPT (Hu et al., 2023) fine-tunes a VLM on high-quality captions of aerial images and introduces a benchmark for remote sensing captioning and visual question answering. GeoChat (Kuckreja et al., 2024) extends LLaVA with region-level inputs and outputs, supporting grounded conversations on high-resolution images. EarthGPT (Zhang et al., 2024) handles optical, SAR and infrared imagery in a single model, and LHRS-Bot (Muhtar et al., 2024) exploits volunteered geographic information to align large amounts of remote sensing images with text.

These models show that VLMs can interpret Earth observation imagery. However, they reason on a single image and on the knowledge acquired during training: they do not retrieve auxiliary information, such as climate records or agricultural statistics, which is necessary for decisions like crop selection.

2.3 Agentic systems for Earth observation

LLM-based agents use a language model to decide, at each step, which action to take, observe its result and plan the following step (Wang et al., 2024a). ReAct (Yao et al., 2023) interleaves reasoning traces with calls to external tools, a pattern now common in agents that interact with search engines, databases or APIs.

In remote sensing, Change-Agent (Liu et al., 2024) combines a change interpretation model with an LLM that invokes it as a tool to answer user requests about bi-temporal images. RS-Agent (Xu et al., 2024) uses an LLM as a controller that selects among remote sensing tools and knowledge sources to solve user queries. GeoLLM-Engine (Singh et al., 2024) provides an environment with geospatial APIs and a large set of tasks for building and evaluating geospatial copilots.

These systems demonstrate that tool use extends the capabilities of language models in Earth observation. However, they focus on image analysis tasks and are evaluated on whether tools are invoked correctly, rather than on the real-world outcome of a recommendation.

2.4 Crop recommendation and AI for agriculture

Crop recommendation is traditionally formulated as supervised classification: given the soil properties of a field, and possibly its climatic conditions, a model predicts the most suitable crop. Ensembles of classical learners trained on soil data are a typical example (Pudumalar et al., 2017). These systems require structured, field-level inputs collected in advance, provide no explanation for their output, and do not exploit imagery. Language models have recently been evaluated on agricultural knowledge, showing that general-purpose models can answer agronomy exam questions with good accuracy (Silva et al., 2023).

2.5 Crawling and retrieval-augmented generation

Web crawlers download resources by systematically traversing the links between them. A crawler maintains a frontier of resources to visit, a scheduling policy that decides the order of visits, and a politeness policy that limits the load imposed on each host (Olston and Najork, 2010). Focused crawlers prioritise the resources most relevant to a given topic, using the content of already visited pages to guide the

exploration (Chakrabarti et al., 1999). These principles are not specific to the Web: any collection of resources connected by relationships can be explored in the same way.

Retrieval-augmented generation (RAG) (Lewis et al., 2020) grounds the output of a language model in documents retrieved for each query and added to the model input. Although originally proposed for text corpora, the same principle applies to structured sources, such as statistical databases, whose records can be retrieved and serialised in the prompt. In this thesis, crawling and retrieval provide the evidence on which the VLM bases its recommendation.

2.6 Evaluation of natural-language outputs

Evaluating free-form answers is an open problem. Human evaluation by domain experts is the most reliable approach, but it is expensive and subjective. Reference-based metrics compare the output with a gold answer, which is not available for advisory tasks. LLM-as-a-judge approaches use a strong language model to score answers and correlate well with human preferences on general tasks (Zheng et al., 2023), but they inherit the biases of the judge and do not verify whether a recommendation is correct in the real world. For crop advisory an objective signal is available: the yield actually obtained by each crop in the following season. This thesis exploits this signal to define an automatic evaluation protocol (Chapter 4).

2.7 Positioning of this work

Table 2.1 compares the proposed system with the main families of related work. To the best of our knowledge, no previous work combines multimodal input, tool-based data collection, climate and agricultural statistics, natural-language explanations and an evaluation based on real outcomes for crop advisory.

Work	**Multi-modal input**	**Tool use**	**Agri-cultural task**	**Climate data**	**NL expla-nation**	**Outcome-based evalua-tion**
CNN/GNN scene classification Han et al. (2017); Zhao et al. (2025)	–	–	Partial	–	–	–
RS foundation models Xiao et al. (2025)	Yes	–	Partial	–	–	–
RS VLMs Hu et al. (2023); Kuckreja et al. (2024); Zhang et al. (2024); Muhtar et al. (2024)	Yes	–	–	–	Yes	–
EO agents Liu et al. (2024); Xu et al. (2024); Singh et al. (2024)	Yes	Yes	–	–	Yes	–
Crop recommenda-tion Pudumalar et al. (2017)	–	–	Yes	–	–	–
LLMs for agronomy Silva et al. (2023)	–	–	Yes	–	Yes	–
This work	Yes	Yes	Yes	Yes	Yes	Yes

Table 2.1: Comparison between this work and the main families of related work.

2.8 Limitations of current approaches

The works reviewed above leave several gaps with respect to crop advisory:

- perception models, including foundation models, solve isolated low-level tasks such as scene classification or image enhancement, but do not answer goal-oriented questions;
- remote sensing VLMs reason on a single image and on their pretrained knowledge, without retrieving the auxiliary data (land cover, climate, yields) on which an agronomic decision depends;
- agentic systems for Earth observation use tools to analyse images, but have not been applied to advisory tasks that require integrating heterogeneous, non-image sources;
- crop recommendation systems require structured field-level inputs and provide no explanation;
- evaluation relies on expert judgement or on benchmarks, rather than on the real-world outcome of the recommendation.

2.9 Research questions and proposed solution

This thesis investigates whether an agentic system that collects multimodal data on demand and reasons over them with a VLM can provide useful crop recommendations, and whether such recommendations can be evaluated automatically. The main research question is:

> *Can an agentic workflow, in which data are collected according to the goal and integrated by a Vision-Language Model, produce explainable crop recommendations that can be evaluated without manual expert assessment?*

It is decomposed into three measurable sub-questions, which are answered in Chapter 5:

- **RQ1**: can a VLM-based agentic pipeline recommend crops whose subsequent yield is above the reference average?
- **RQ2**: how do an open-weight 7B VLM and a proprietary VLM compare on this task?
- **RQ3**: how sensitive is the evaluation to using NASS forecasts instead of final yields as ground truth?

To answer these questions, this work proposes a system composed of an agentic crawler and a VLM. The crawler explores a geographic grid

over the contiguous United States of America and, for each location, decides which data to retrieve and from which source: satellite imagery, land-cover maps, climate variables and agricultural statistics. The abundance of open-access reports about the yield across US cropland is the main reason for choosing this geographic area; however the application can be generalized to every other land. The VLM receives the integrated evidence for the requested location and produces a recommendation of a main crop and a rotation crop, together with an explanation in natural language. The recommendations are evaluated automatically against the yields observed in the following season. The system is described in Chapter 3.

Chapter 3

The Proposed System

This chapter describes the proposed system. Section 3.1 formalises the task and Section 3.2 gives an overview of the architecture. The following sections describe its components: the geographic grid on which data collection is organised (Section 3.3), the agentic crawler (Section 3.4), the multimodal storage (Section 3.5), the integration of the collected data into the prompt (Section 3.6) and the VLM that produces the recommendation (Section 3.7).

3.1 Problem formulation

Let $p = (\varphi, \lambda)$ be a point, expressed as latitude and longitude, in the contiguous United States, and let $B(p)$ be the 10×10 km bounding box centred on p. Given p, the system returns a triple $(c_{\text{main}}, c_{\text{sec}}, e)$, where c_{main} is the crop recommended as the main crop for $B(p)$, c_{sec} is a

secondary crop suitable for rotation with c_{main}, and e is an explanation in natural language.

To produce this answer, the system collects a set of evidence

$$\mathcal{D}(p) = \{\, I(p),\ L(p),\ C(p),\ Y(p) \,\}, \tag{3.1}$$

where $I(p)$ is an RGB Sentinel-2 image of $B(p)$, $L(p)$ is the land-cover composition of $B(p)$ derived from the Cropland Data Layer (CDL), $C(p)$ is a set of climate variables from ERA5 and several indexes computed on GeoTIFF images, and $Y(p)$ contains the historical yields of the relevant crops from USDA Quick Stats. The content of $Y(p)$ depends on $L(p)$: yields are retrieved only for the crops that are actually grown in the area. The data sources are described in detail in Chapter 4.

3.2 System overview

Figure 3.1 shows the architecture of the system, which is composed of four main components:

- a **geographic grid** that discretises the territory into locations to be visited;
- an **agentic crawler** that visits the grid and create a download plan in order to collect data for each location;
- a **multimodal storage system** that archives heterogeneous

data in a flexible structure;

- a **Vision-Language Model** that integrates the collected data and produces the recommendation.

Figure 3.1 shows an overview of the proposed system. In particular for the architecture the external components are reusable, the Agentic Crawler and the VLM module have been re-engineered, and both the geographic grid and the Multimodal Storage System are specifically designed and implemented for this work. The system operates in two phases. In the *collection phase*, the grid is built using the CDL available on Google Earth Engine, and the agentic crawler traverses it, retrieving the data of each node through Earth Engine and storing them: structured data in ArangoDB and images in Google Drive. In the *inference phase*, given an input location, the system retrieves the data of the corresponding node from the storage, queries USDA Quick Stats for the yields of the crops grown in the area (dashed arrow in Figure 3.1), builds the prompt and submits it to the VLM, which returns the recommendation.

3.3 Geographic grid and crawl graph

The territory of the contiguous United States is discretised with a regular grid with a spacing of 0.5° in latitude and longitude, which corresponds to approximately 55 km in the north–south direction and

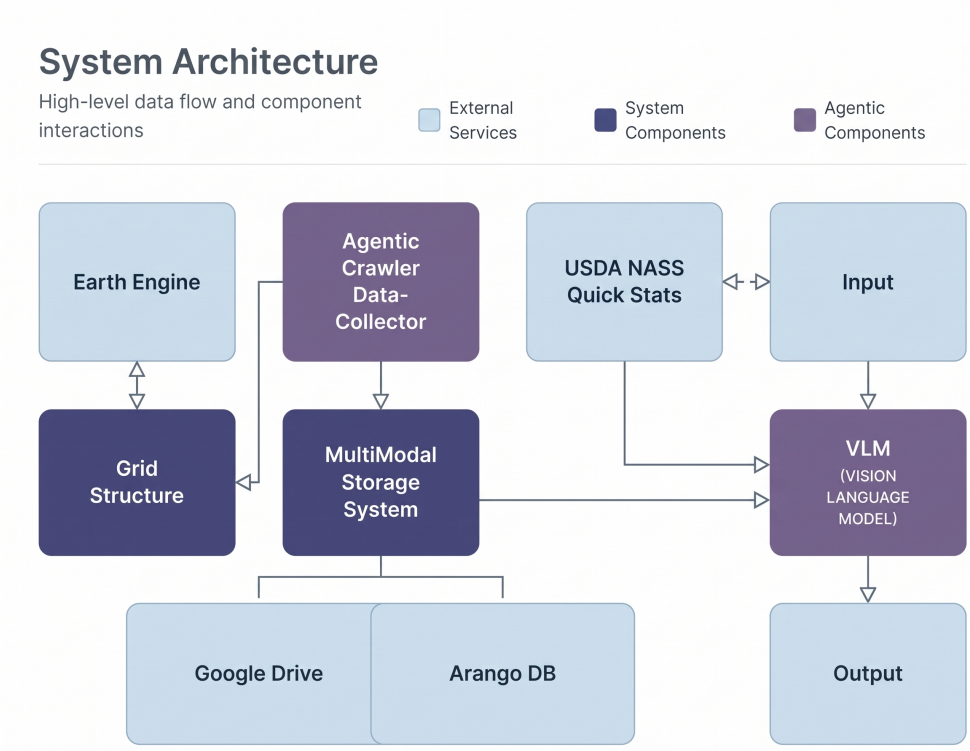


Figure 3.1: Architecture of the proposed system. External services are shown in light blue, system components in dark blue and agentic components in purple.

40–47 km in the east–west direction at US latitudes. For each grid point p, a 10×10 km bounding box $B(p)$ centred on the point is considered.

Since most of the territory is not used for agriculture, the grid is filtered according to the cropland fraction of each box, computed from the CDL as

$$f(p) = \frac{|\{\, x \in B(p) : \mathrm{CDL}(x) \in \mathcal{K} \,\}|}{|B(p)|}, \tag{3.2}$$

where x ranges over the pixels of $B(p)$ and $\mathcal{K}$ is the set of CDL classes corresponding to cultivated crops. Only the 100 boxes with the highest cropland fraction are retained. The retained points form the nodes V of the crawl graph $G = (V, E)$. Two nodes are connected by an edge, so that the topology of the graph reflects the geography of the territory, in the same way as hyperlinks define the topology of the Web.

3.4 Agentic crawler

3.4.1 From rule-based to agentic crawling

A traditional crawler (Figure 3.2) is driven by static, rule-based policies. A sequential spider module extracts the next resource from a queue, a download module retrieves it through an HTTP (Hypertext Transfer Protocol) request, and a storage module saves the result and reports

the processing status back to the spider. The order of the visits and the content to be downloaded are fixed in advance.

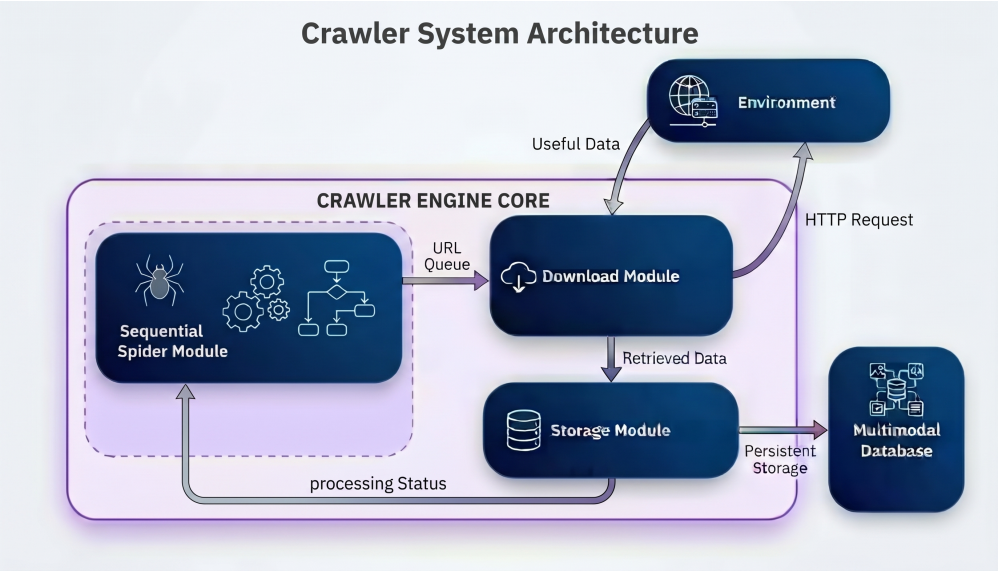


Figure 3.2: Architecture of a traditional rule-based crawler.

In the proposed agentic crawler (Figure 3.3), a reasoning module is added before the spider module. For each node, the reasoning module receives from the storage the current state of the node, that is, which data have already been collected and with which quality, and decides which data are still needed to satisfy the goal (*target data*). The spider module translates this decision into a download plan, which specifies the tools to invoke and their parameters, and schedules it according to the politeness constraints. The download module invokes the tools on the environment, here the external data providers, and the retrieved data are saved by the storage module, which updates the state of the node. The loop is repeated until the goal is satisfied for every node.

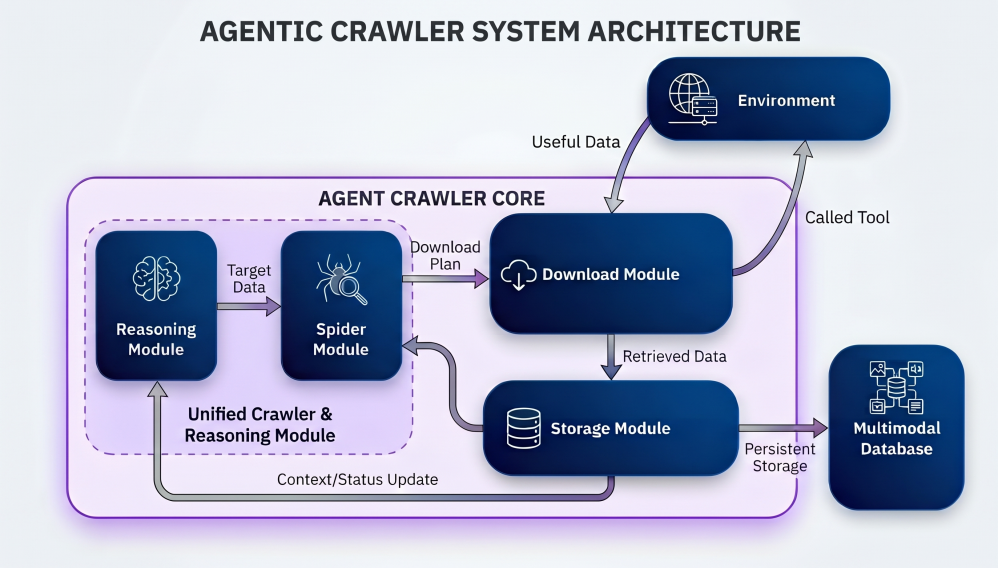


Figure 3.3: Architecture of the proposed agentic crawler. The reasoning module decides which data to collect based on the goal and on the state of each node.

3.4.2 Tools

The tools available to the crawler are listed in Table 3.1. Each tool wraps a call to an external service and returns data in a uniform format, so that new sources can be added without modifying the rest of the system.

Tool	Input	Output	Source
`download_sentinel`	coordinates, bounding box, time window, max. cloud cover, bands, scale	GeoTIFF	GEE
`download_NASS_CDL`	coordinates, bounding box, year, scale	Histogram of CDL class	GEE
`extract_era5`	bounding box, time window, variables	aggregated climate variables	GEE
`Generate_report`	GeoTIFF	Vegetation index report	One Drive
`query_quickstats`	state, crop, year	yields and forecasts	NASS API

Table 3.1: Tools available to the agentic crawler.

3.4.3 Decisions of the reasoning module

The reasoning module makes the following decisions trough a rule based approach:

- **node selection**: which node to visit next, giving priority to nodes with a higher cropland fraction;
- **data selection**: which sources are still missing for the current node, so that already available data are not downloaded again;
- **quality control**: whether the retrieved data are adequate, for example whether the cloud cover of the image exceeds a threshold, in which case a different time window is requested;

3.5 Multimodal storage

The storage system separates structured data from images. Structured data are stored in ArangoDB, a multi-model database that supports both documents and graphs: nodes and edges of the crawl graph are stored in two collections, and each node is a JSON document that can be extended with new fields as the crawler collects new information, without redefining the schema. Images are exported as GeoTIFF files to Google Drive, and each node stores a reference to its image. Table 3.2 shows the structure of a node document.

```
{
        "State": "Texas",
        "coordinates": [
            -97.764443486,
            26.4348793533
        ],
        "climate_data": {
            "leaf_area_index_high_vegetation": 0,
            "leaf_area_index_low_vegetation": 1.4397742462,
            "..."
        },
        "histogram": {
            "Background": 0.01,
            "Corn": 4.94,
            "Cotton": 30.64,
            "Rice": 0.02,
            "..."
        },
        "indexes": {
            "NDVI": 0.399185871,
            "NDWI": -0.4599860306,
            "..."
        }
    }
```


Table 3.2: Structure of a node document in ArangoDB (simplified).

3.6 Adaptive data integration and prompt construction

At inference time, the system builds the input of the VLM from the data of the node closest to the requested location. Integration is adaptive: the land-cover composition $L(p)$ determines which crops are relevant, and only for these crops the yields of the previous season are retrieved from Quick Stats. This step is a form of retrieval-augmented generation over structured data: the records retrieved for the query are serialised and added to the prompt, so that the answer of the model is grounded in data specific to the location rather than only in its pretrained knowledge.

The prompt is composed of four parts:

1. a **system prompt** that defines the role of the model, the task (recommending a main crop and a rotation crop) and the required structure of the answer;
2. the **image** $I(p)$, converted from GeoTIFF to a JPEG-compressed RGB image;
3. a **land analysis** JSON object containing $L(p)$ and $C(p)$;
4. a **yield** JSON object containing $Y(p)$.

3.7 VLM-based reasoning

The VLM is the component that integrates the evidence and produces the recommendation. Its architecture, shown in Figure 3.4, follows the common design of VLMs: a visual encoder converts the image into a sequence of visual tokens, a connector projects them into the embedding space of the language model, and the language model processes them together with the textual tokens of the prompt to generate the answer.

The VLM is decoupled from data collection: it does not know which tools were used or in which order, and receives only the integrated evidence and the description of the task. This separation has two advantages. First, different VLMs can be compared on exactly the same input, as done in Chapter 5. Second, the data sources can be

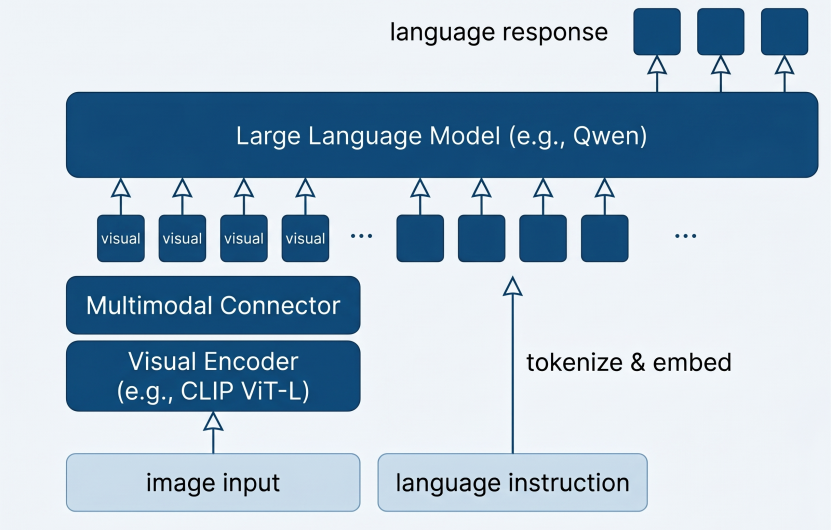


Figure 3.4: General architecture of a Vision-Language Model (adapted from (Liu et al., 2023)).

changed without modifying the model or the prompt, as long as the data are serialised in the same format.

The system prompt asks the model to state the recommended main crop and rotation crop explicitly and to justify them with reference to the image, the land cover, the climate and the yields. The two crops are then manually extracted from the answer. When the answer mentions more than one candidate, the first crop proposed for each role is retained.

Chapter 4

Experimental Setup

This chapter describes the experimental setup used to evaluate the proposed system. Section 4.1 presents the implementation and the computing environment, Section 4.2 the data sources, and Section 4.3 the locations selected for the evaluation. Sections 4.4 and 4.5 define the ground truth and the evaluation metrics, and Section 4.6 describes the two VLMs compared and their configuration.

4.1 Implementation and environment

The pipeline is implemented in Python 3.11.9, relying on **PyTorch**, **Transformers**, and **scikit-learn** for the agentic and reasoning modules, **Pillow** and **Rasterio** for geospatial and raster image processing, and the Earth Engine Python API (**ee**) alongside **requests** for querying GEE and the USDA Quick Stats database. All satellite and climate

data are accessed through Google Earth Engine (GEE) (Gorelick et al., 2017), a cloud platform for geospatial analysis that hosts a large public catalogue of satellite imagery and derived products. The download module submits export tasks that are executed on Google's infrastructure, which distributes the computational load and avoids downloading raw data locally; the resulting images are exported to Google Drive as GeoTIFF files.

Structured data and the crawl graph are stored in ArangoDB (Weinberger and Celler), which runs in a Docker container. The container isolates the database from the rest of the code and makes the environment reproducible on different machines.

Qwen2-VL is executed locally on two GPUs, one RTX 3090 24GB Ram, and one RTX 5000 Ada 32GB RAM with no quantization, while Gemini is accessed through the Google AI Studio API.

4.2 Data sources

Table 4.1 summarises the data sources used in this work and their role in the system.

4.2.1 Sentinel-2

Sentinel-2 is a European Space Agency mission, part of the Copernicus programme, composed of two satellites that acquire multispectral

Source	Content used	Resolution	Period	Role
Sentinel-2 L2A (harmonized)	Spectral Bands	10 m	2023	image input
CDL	crop-specific land cover	30 m	2023	cropland filter, land-cover composition
ERA5	climate variables (Table 4.3)	Buffer_size	2023	climate descriptors
USDA Quick Stats	yields and yield forecasts	state	2024–2025	model input, ground truth

Table 4.1: Data sources used in this work.

images in 13 bands with a revisit time of five days at the equator (Drusch et al., 2012). This work uses the harmonized Level-2A collection available on GEE (Union/ESA/Copernicus), which provides surface reflectance corrected for atmospheric effects. Only the red, green and blue bands, at 10 m resolution, are used to build the RGB image passed to the VLM. Each scene includes cloud cover metadata, which are used to discard cloudy acquisitions. In order to compute the vegetation index scores, additional bands like Near Infra-red (NIR) were acquired, as shown in Table 4.2.

Index	Bands
Normalized Difference Vegetation Index	NIR, RED
Normalized Difference Water Index	NIR, GREEN
Chlorophyll Index Green	NIR,GREEN

Table 4.2: ERA5 variables used as climate descriptors.

4.2.2 Cropland Data Layer

The Cropland Data Layer (CDL) is a land-cover map of the contiguous United States produced every year by the USDA National Agricultural Statistics Service (NASS), which assigns each 30 m pixel to a crop type or to another land-cover class (Boryan et al., 2011). In this work the CDL is used to compute the cropland fraction of each grid cell (Section 3.3) and the land-cover composition of each bounding box, that is, the fraction of the area occupied by each crop.

4.2.3 ERA5

ERA5 is the climate reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), which combines numerical weather models with observations to provide consistent, gridded estimates of atmospheric and land-surface variables from 1940 onwards (Hersbach et al., 2020). The variables used in this work are listed in Table 4.3.

Variable	**Aggregation**
Leaf area index	average/monthly
Volumetric soil water (layer 1)	average/monthly
Surface net solar radiation	average/monthly
2 m air temperature	average/monthly
Total precipitation	average/monthly

Table 4.3: ERA5 variables used as climate descriptors.

4.2.4 USDA Quick Stats

Quick Stats[1] is the database through which NASS publishes US agricultural statistics, queryable by commodity, location and period (NASS). This work uses the state-level yield of each crop, both as input to the model and as ground truth (Section 4.4). NASS also publishes yield forecasts during the season, based on farmer surveys and field observations; these forecasts are used as an alternative ground truth.

4.3 Evaluation points

The evaluation is performed on twelve locations, selected among the 100 retained grid cells as a compromise between geographic variety and richness of crop cover. Each location belongs to a different state, so that the evaluation covers different climates and agricultural systems, from the Corn Belt to the semi-arid Great Plains and the irrigated Central Valley of California. For each state, the node with the greatest cropland cover was chosen. The locations are listed in Table 4.4.

4.4 Ground truth

The evaluation exploits the yields actually obtained in the season following the one whose data are given to the model. Let $Y_{c,s,t}$ be the yield of crop c in state s in year t, and $\bar{Y}_{c,t}$ the corresponding national

[1] `https://quickstats.nass.usda.gov/`

State	Latitude	Longitude	Cropland fraction
Arkansas	35.93	-90.26	93.22%
California	36.43	-120.26	94.07%
Colorado	38.43	-102.26	86.44%
Illinois	40.93	-88.26	92.45%
Iowa	43.43	-94.26	91.93%
Kansas	38.43	-100.76	92.65%
Minnesota	47.93	-96.76	95.08%
Mississippi	33.43	-90.76	86.2%
Missouri	36.43	-89.76	91.68%
Nebraska	42.43	-97.26	91.26%
North Dakota	46.93	-97.26	89.8%
Texas	33.43	-101.76	88.75%

Table 4.4: Locations used for the evaluation.

average. The *relative yield*

$$r_{c,s,t} = 100 \cdot \frac{Y_{c,s,t}}{\bar{Y}_{c,t}} \tag{4.1}$$

expresses the yield of a crop in a state as a percentage of the national average, and makes yields of different crops comparable.

The model receives the yields of year $T - 1 = 2024$, and its recommendations are evaluated on year $T = 2025$. Two ground truths are considered:

- GT_A: the final yields published by NASS for year T;
- GT_B: the NASS yield forecasts for year T on the earliest available month.

Comparing the two measures how sensitive the evaluation is to the source of the ground truth, which is relevant in practice because fore-

casts are available months before final yields. When the forecast is not available for a crop, the corresponding score is marked as not available (N/A).

4.5 Evaluation metrics

For each location p in state s, with recommended main crop m and secondary crop n, three binary scores are computed.

Main crop suitability. The main crop is considered suitable if its yield in year T is above the national average:

$$S_{\text{main}}(p) = \mathbf{1}\left[\, r_{m,s,T} > 100 \,\right]. \tag{4.2}$$

Secondary crop suitability. Similarly, for the secondary crop:

$$S_{\text{sec}}(p) = \mathbf{1}\left[\, r_{n,s,T} > 100 \,\right]. \tag{4.3}$$

Rotation suitability. A rotation is considered suitable if the gap between the relative yields of the two crops decreases from year $T-1$ to year T, that is, if the secondary crop performs increasingly close to the main crop or if the secondary crop has a higher suitability expressed as previously explained:

$$S_{\text{rot}}(p) = \begin{cases} \mathbf{1}\,[\,|\Delta_T| < |\Delta_{T-1}|\,] & \text{if } r_{n,s,T} < r_{m,s,T} \\ \mathbf{0}\,[\,|\Delta_T| > |\Delta_{T-1}|\,] & \text{if } r_{n,s,T} < r_{m,s,T} \\ 1 & \text{if } r_{n,s,T} > r_{m,s,T} \end{cases}$$

We must take into account that rotation serves as an alternative rather than a substitute; because of this, we can rely on a simple gap reduction metric. The accuracy of a model on a metric is the fraction of locations with score 1, computed over the n locations for which the score is available:

$$\text{Acc} = \frac{1}{n} \sum_{p} S(p). \tag{4.4}$$

Since the number of locations is small, the uncertainty of the accuracy is estimated with the bootstrap (Efron and Tibshirani, 1993): the locations are resampled with replacement $B = 1000$ times, the accuracy is computed on each resample, and the 2.5th and 97.5th percentiles of the resulting distribution are reported as standard error and 95% confidence interval.

4.6 Models and inference configuration

Two VLMs are compared: an open-weight model that can be run on local hardware and a proprietary model accessed through an API.

Qwen2-VL-7B-Instruct. Qwen2-VL (Wang et al., 2024b) is a family of open-weight VLMs developed by Alibaba Cloud. It combines a Vision Transformer encoder with a Qwen2 language model and processes images at their native resolution, mapping them to a variable number of visual tokens. This work uses the instruction-tuned 7B version (Alibaba, 2024), chosen because it fits on the available hardware.

Gemini 3.8 Flash. Gemini 3.8 Flash (Google) is a proprietary multimodal model developed by Google, which accepts images and text as input and supports a long context window and configurable reasoning (*thinking*) levels.

The two models differ considerably in size and training resources, so the comparison does not aim to establish which architecture is better, but to assess whether a small open model that can be deployed locally achieves results comparable to a frontier proprietary model on this task. Both models receive exactly the same prompt for each location. The inference parameters are reported in Table 4.5.

Parameter	**Qwen2-VL-7B-Instruct**	**Gemini 3.8 Flash**
Access	local weights	API
Temperature	0.0 (Default)	1.0 (Default)
Max output tokens	2048	65,536
Thinking level	–	Default
Image resolution	2541 x 2004 pixel	2541 x 2004 pixel
Runs per location	1	1

Table 4.5: Inference configuration of the two models.

4.7 Evaluation Strategies

For each point, a structured question was submitted only once. The structure of the input was kept identical; however, different locations were included. The evaluation was designed to assess both qualitative and quantitative results. The qualitative measure considers how well-structured the response is and whether it is free from hallucinations, while domain expert review would offer an even more rigorous benchmark. Regarding the quantitative results, the explicit crop recommendation was extracted from each response and evaluated against both observed and forecast ground truths using the metrics defined above. This procedure was carried out for both VLMs and produced four tables: the first two show the binary scores and the average of each crop recommended for the first and second model, respectively; the third shows a direct comparison between the two models on the forecast ground truth; and the fourth summarizes the aggregated metrics.

Chapter 5

Results and Discussion

This chapter presents the results of the evaluation. Section 5.1 reports the quantitative results, Section 5.2 compares two example responses, and Section 5.3 analyses the errors observed in them. Section 5.4 discusses the results with respect to the research questions.

5.1 Quantitative results

Tables 5.1 and 5.2 report, for each evaluation point, the crops recommended by the two models, their relative yields in 2024 and 2025 and the resulting scores with the final yields (GT_A). Table 5.3 reports the scores obtained with the NASS forecasts (GT_B). Table 5.4 and Figure 5.1 summarise the accuracy of the two models on the three metrics.

		2024		2025				
State	**Main → Secondary**	**Main**	**Sec.**	**Main**	**Sec.**	S_{main}	S_{sec}	S_{rot}
Texas	Cotton → Winter wheat	79.4	61.1	74.9	72.9	0	0	1
North Dakota	Spring wheat → Canola	128.2	100.5	119.5	114.6	1	1	1
Illinois	Corn → Soybeans	122.4	128.0	120.7	125.0	1	1	1
Arkansas	Cotton → Soybeans	159.6	108.6	147.5	108.6	1	1	1
Minnesota	Corn → Soybeans	98.1	94.8	113.3	102.7	1	1	0
Mississippi	Soybeans → Corn	110.7	105.0	110.7	101.5	1	1	0
Iowa	Corn → Soybeans	119.0	118.5	118.4	125.5	1	1	1
Nebraska	Corn → Soybeans	106.0	113.6	109.0	128.4	1	1	1
Colorado	Winter wheat → Sorghum	69.0	46.1	78.9	80.0	0	0	1
California	Rice → Corn	129.6	102.6	128.2	97.6	1	0	0
Kansas	Sorghum → Winter wheat	100.0	84.8	138.4	99.4	1	0	0
Missouri	Soybeans → Corn	96.8	103.3	98.8	104.3	0	1	1

Table 5.1: Recommendations of Qwen2-VL-7B-Instruct, relative yields (% of the national average) of the recommended crops, and scores computed with the final yields (GT_A).

		2024		2025				
State	**Main → Secondary**	**Main**	**Sec.**	**Main**	**Sec.**	S_{main}	S_{sec}	S_{rot}
Texas	Cotton → Winter wheat	79.4	61.1	74.9	72.9	0	0	1
North Dakota	Spring wheat → Canola	128.2	100.5	119.5	114.6	1	1	1
Illinois	Corn → Soybeans	122.4	128.0	120.7	125.0	1	1	1
Arkansas	Cotton → Peanuts	159.6	142.0	147.5	128.6	1	1	0
Minnesota	Soybeans → Spring wheat	94.8	149.0	102.7	150.0	1	1	1
Mississippi	Soybeans → Corn	110.7	105.0	110.7	101.5	1	1	0
Iowa	Corn → Soybeans	119.0	118.5	118.4	125.5	1	1	1
Nebraska	Corn → Soybeans	106.0	113.6	109.0	128.4	1	1	1
Colorado	Winter wheat → Millet	69.0	113.0	78.9	122.0	0	1	1
California	Rice → Sunflowers	129.6	87.0	128.2	81.0	1	0	0
Kansas	Winter wheat → Sorghum	84.8	100.0	99.4	138.4	0	1	1
Missouri	Soybeans → Corn	96.8	103.3	98.8	104.3	0	1	1

Table 5.2: Recommendations of Gemini 3.8 Flash, relative yields (% of the national average) of the recommended crops, and scores computed with the final yields (GT_A).

State	Qwen2-VL			Gemini		
	S_{main}	S_{sec}	S_{rot}	S_{main}	S_{sec}	S_{rot}
Texas	0	0	0	0	0	0
North Dakota	N/A	1	N/A	N/A	1	N/A
Illinois	1	1	1	1	1	1
Arkansas	1	1	1	1	1	0
Minnesota	1	1	0	1	1	1
Mississippi	1	1	1	1	1	1
Iowa	1	1	1	1	1	1
Nebraska	1	1	1	1	1	1
Colorado	0	0	1	0	N/A	N/A
California	1	N/A	N/A	1	N/A	N/A
Kansas	1	1	0	1	1	1
Missouri	1	1	1	1	1	1

Table 5.3: Scores of the two models computed with the NASS yield forecasts (GT_B). N/A: forecast not available for at least one of the crops involved.

Metric	Model	GT_A (final yields)		GT_B (forecasts)	
		Accuracy	95% CI	Accuracy	95% CI
Main crop	Qwen2-VL	75.0% (9/12)	[50.0, 91.7]	75.0% (9/12)	[50.0, 100.0]
	Gemini	66.7% (8/12)	[41.7, 91.7]	75.0% (9/12)	[50.0, 100.0]
Secondary crop	Qwen2-VL	66.7% (8/12)	[41.7, 91.7]	72.7% (8/11)	[45.5, 100.0]
	Gemini	83.3% (10/12)	[58.3, 100.0]	75.0% (9/12)	[50.0, 100.0]
Rotation	Qwen2-VL	66.7% (8/12)	[41.7, 91.7]	70.0% (7/10)	[40.0, 90.0]
	Gemini	75.0% (9/12)	[50.0, 100.0]	77.7% (7/9)	[40.0, 100.0]

Table 5.4: Accuracy of the two models with the two ground truths. Confidence intervals are bootstrap percentile intervals (1,000 resamples). Bootstrap standard errors range from 0.11 to 0.16.

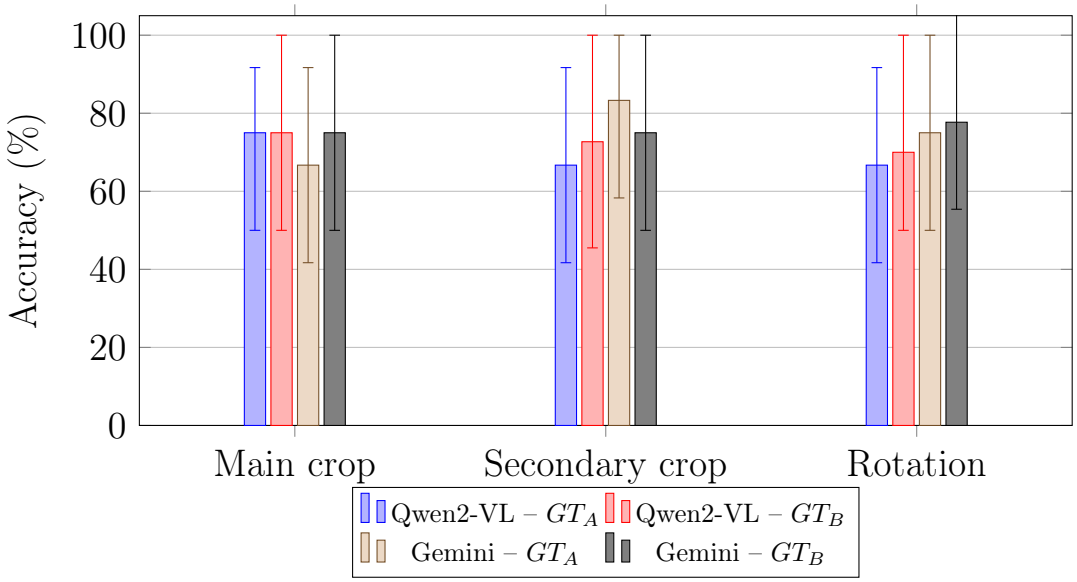


Figure 5.1: Accuracy of the two models with the two ground truths. Error bars show the 95% bootstrap confidence intervals.

5.2 Qualitative analysis

To illustrate the differences between the two models, this section reports their answers for the location in California, where both models recommend rice as the main crop.

Qwen2-VL-7B-Instruct – California

Based on the provided data, the best crop to plant in the specified location in California is Rice. Here's the reasoning:

1. Historical Yield Data: The historical yield data for rice in the specified area shows consistent yields ranging from 8,290 to 8,830 pounds per acre. This indicates that rice is a reliable crop with a proven yield in the region.

2. Optical Imagery: The optical image shows a large expanse of green fields, which is characteristic of rice fields. The uniformity and density of the green patches suggest that rice is a suitable crop for this area.

3. Soil Information: The soil in the area is suitable for rice cultivation, as indicated by the high leaf area index (LAI) and the volumetric soil water layer 1 values.
4. Climate Data: The climate data shows high leaf area index (LAI) and surface net solar radiation, which are favorable for rice growth. The temperature and precipitation levels are also suitable for rice cultivation.
5. Crop Rotation: A suitable crop rotation for rice could include: first year: rice; second year: fallow, or a cover crop (such as alfalfa or clover) to improve soil health and fertility; third year: rice again or another suitable crop like corn or soybeans.
This rotation helps maintain soil health and prevents the buildup of pests and diseases that can occur with continuous rice planting.

Gemini 3.8 Flash – California

Due to the heavy clay soils and flood-irrigation infrastructure, conventional crop rotations can be challenging. However, continuous monoculture can intensify issues with herbicide-resistant aquatic weeds (e.g., California arrowhead, watergrass) and rice blast (*Magnaporthe oryzae*). The following rotation and conservation system is recommended:

Years	**Season**	**Crop / practice**	**Purpose**
1–3	Summer (Apr–Sep)	Rice (flooded culture)	Peak commercial grain production using high-yielding California medium-grain varieties (e.g., M-206, M-209).
	Winter (Oct–Feb)	Post-harvest winter flooding	Stubble decomposition, waterfowl foraging (Pacific Flyway integration), and suppression of winter weed growth.
4	Spring–Fall	Fallow / dry-tilled fallow (or safflower/sunflower)	Deep soil cracking (vertisol aeration), thorough mechanical weed seedbank depletion, and conservation of irrigation allocations during dry/drought years.
5 (optional)	Fall–Spring	Winter wheat / triticale	Shallow root system development, erosion control, and weed cycle disruption prior to re-entering flooded rice production.

Management tips:

- Water and straw management: incorporate rice straw in early autumn followed by shallow winter flooding instead of burning, taking advantage of local working lands conservation programs.
- Nutrient management: band pre-plant ammoniacal nitrogen into dry soil before the flood to minimize volatilization and maximize nitrogen use efficiency in heavy clay basins.

The two answers differ in focus. Qwen2-VL concentrates on justifying the main crop, referring explicitly to each part of the input (yields, image, soil and climate variables), while the rotation is described only briefly. Gemini takes the main crop for granted and devotes most of the answer to an agronomically detailed rotation and management plan, drawing extensively on knowledge acquired during pre-training rather than on the input data. This difference is consistent with the quantitative results: Gemini obtains a higher accuracy on the secondary crop, while Qwen2-VL obtains a higher accuracy on the main crop.

5.3 Error analysis

The answer of Qwen2-VL contains statements that are not supported by the input, which illustrate typical failure modes of VLMs in this domain:

- **misinterpretation of variables**: the leaf area index (LAI) is cited as evidence of soil suitability, although it describes the

density of the vegetation canopy and not the soil; the same variable is then cited again among the climate data;

- **unverifiable visual claims**: the model states that the image shows green fields "characteristic of rice fields", but rice is difficult to distinguish reliably from other green crops in a single RGB image at 10 m resolution, and the claim may be driven by the yield data rather than by the image;

- **vague recommendations**: the rotation includes alternatives (fallow or cover crop, rice or corn or soybeans) instead of a single secondary crop, which makes the extraction of the recommendation ambiguous.

These errors suggest that the image contributes less to the decision than the structured data, a hypothesis that could be tested with an ablation study in which the image is removed from the prompt. On the other hand, Gemini maintains a coherent reasoning process, showing a higher focus on the resolution of the task without introducing hallucinations, thereby achieving better reliability.

5.4 Discussion

RQ1: suitability of the recommendations. Both models recommend a main crop whose 2025 yield is above the national average in most locations: 9 out of 12 for Qwen2-VL and 8 out of 12 for Gemini.

The failures are concentrated in two groups. In Texas and Colorado, the recommended crops are below the national average for both models; these states are largely semi-arid, and most crops grown there have yields below the national average regardless of the recommendation. Since the metric compares each state with the national average, these locations are penalised by construction: this is a bias of the metric rather than necessarily an error of the model, and normalising yields by the historical average of each state would remove it. The remaining failures are marginal: Missouri soybeans (98.8%) for both models, and Kansas winter wheat (99.4%) for Gemini. Such cases show that a binary threshold at 100% is sensitive to small variations, and that a continuous measure, such as the average relative yield of the recommended crops, would complement the accuracy.

Regarding crop rotation, Qwen2-VL achieved an accuracy of 66.7%, while Gemini reached 75.0% on the final yields. Given the sample size of twelve locations, the former score is only slightly better than random chance. Since the rotation metric serves only as a weak proxy for agronomic suitability, these rotation results should be interpreted with caution. Furthermore, Gemini's higher rotation score may be correlated with its lower performance on the main crop, which indirectly benefited the rotation metric; therefore, the same cautious conclusion applies.

RQ2: open-weight versus proprietary model. Qwen2-VL obtains a higher accuracy on the main crop (75.0% versus 66.7%), and

Gemini on the secondary crop (83.3% versus 66.7%). Gemini tends to propose secondary crops different from the dominant ones, such as peanuts in Arkansas, spring wheat in Minnesota and millet in Colorado, which in these cases achieved high relative yields. However, the 95% confidence intervals of the two models overlap widely for all metrics, so none of these differences is statistically significant with the current sample. The main difference between the two models is qualitative, as discussed in Section 5.2.

RQ3: sensitivity to the ground truth. For the main and secondary crops, the accuracies obtained with the NASS forecasts differ from those obtained with the final yields by at most 8.3 percentage points, and the scores agree in most locations. This indicates that forecasts, which are available months before final yields, could be used for a preliminary evaluation. The rotation score is more sensitive, because it depends on the difference between two forecasts and because several forecasts are not available.

Persistence of yields. The model receives the yields of 2024 and is evaluated on those of 2025. Relative yields are strongly autocorrelated: for the main crop recommended by Qwen2-VL, being above or below the national average in 2025 coincides with the 2024 situation in 10 out of 12 states (11 out of 12 for Gemini). Part of the accuracy may therefore derive from the yield data passed in the prompt rather than

from the integration of image, land cover and climate.

Conclusion

The main contributions of this work are threefold. First, we propose an architecture for crop advising based on an agentic crawler operating over a geographic grid, which dynamically adapts data collection to both the task objective and existing local data. Second, we introduce an adaptive integration pipeline that fuses heterogeneous data into a multimodal prompt, where the land-cover composition of each location dictates which agricultural statistics are retrieved, following the principles of structured retrieval-augmented generation. Finally, this work proposes an automated evaluation protocol based on observed yields from the subsequent season, using both final yields and NASS forecasts as alternative ground truths. The evaluation also presents an experimental comparison between an open-weight 7B VLM (Qwen2-VL) and a proprietary model (Gemini 3.8 Flash) across twelve locations in different US states.

Several promising research directions can extend the methodology of this work in terms of both evaluation strategies and architectural design.

About evaluation strategies, the experimental scale and granularity could be enhanced by expanding the evaluation across a larger set of locations and leveraging county-level yields from USDA Quick Stats, which align much closer with the spatial scale of the analyzed agricultural areas. In parallel, the evaluation framework could complement the current binary classification with continuous performance metrics, and define not only a rotation score but also a more structured ranking of which crop could be planted in the area.

Furthermore, a thorough benchmarking strategy should include comparative baselines governed by simple heuristics, such as selecting the historically dominant crop or replicating the top-performing crop from the preceding season, alongside comprehensive ablation studies to isolate and quantify the specific contributions of optical imagery, land-cover distributions, and ERA5 climatic variables.

From an architectural standpoint, a key extension involves transitioning from the current rule-based reasoning module to an autonomous, LLM-driven reasoning controller. Such an evolution would allow the agent to improve its dynamic retrieval plan, its complex multi-step decision chains, and its external tool management based on intermediate findings rather than predefined reasoning logic. This architectural progression could be further enriched by evaluating newer open-weight VLMs, testing models explicitly fine-tuned on Earth observation benchmarks, and quantitatively assessing the semantic quality and factual

rigor of the generated natural language rationales through agronomist validation or LLM-as-a-judge protocols.

Finally, a compelling evolution of the advisory scope lies in transitioning from purely biological yield optimization to economic advisory, recommending crop allocations based on projected profitability and return on investment. Because commodity market prices, supply chain trends, and production costs fluctuate much more rapidly than the physical characteristics of the land, dynamically retrieving and reasoning over economic data streams represents an ideal testbed for assessing the full adaptability of an agentic data collection framework.

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